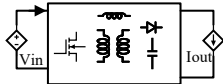


Design Requirements

Power electronic converters



Design Variables

Switching
Frequency

Voltage

Ripple

Current

size

Cost

DRL Agent

Analytical Model

Efficiency Model

- Power Component Loss
- Magnetic Component Loss

Topology Model

- Topological Difference
- Topological Generation

Cost Model

- Core Cost
- Winding Cost

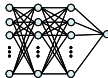
Ripple Model

Dynamic Model

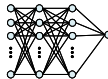
Steady Mode

DRL Model

Critic Network

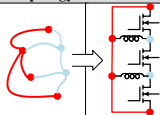


Actor Network

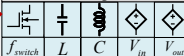


Design objectives

Topology Generation



Parameter Optimization



Multi-Objective Design Optimization

Efficiency

Ripple

Steady & Dynamic